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The effects of velocity loss thresholds during resistance training on lower-limb performance enhancement: A systematic review with meta-analysis

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Abstract

This study assessed the effects of squat training with different velocity loss (VL) thresholds on squat strength, heavy-load squat velocity (HLSV), light-load squat velocity (LLSV), countermovement jump (CMJ) height, and sprint time (ST). A systematic search of electronic databases was conducted. Meta-analysis was used to examine effects of trainings with different VL thresholds and used meta-regression to examine the interaction effects (correlation) between training effects and VL in different performance-based outcomes. Our systematic search yielded 1017 articles, 13 of which were included in the present study. Meta-analysis revealed that resistance training with both low (0%–20%) and high (30%–45%) VL thresholds improved squat strength, HLSV, LLSV, and CMJ height, but only low VL improved ST ($p < 0.05$). Training gains for HLSV, LLSV, CMJ ($p = 0.088$), and ST with low VL were superior to those with high VL ($p < 0.05$). Meta-regression revealed that CMJ ($p = 0.076$) and ST ($p = 0.010$) demonstrated improvements as VL decreased in the range of 0% to 45%. In conclusion, squat resistance training with low (≤20%) VL provides relatively more effective training stimuli, improving performance in power and speed tasks. Optimal VL ranges may be in the range 10% to 20%, ≤20%, and ≤10% for the HLSV and LLSV, CMJ, and ST, respectively.

Keywords

Countermovement jump, power, speed, sprint, squat, strength, velocity-based training

Introduction

Strength training programs are traditionally based on the percentage-based training (PBT) methodology.[1] The selected load and training volume (sets × repetitions) are normally prescribed based on the physical quality (strength, speed-strength, hypertrophy, etc.) that the strength and conditioning coach is attempting to develop.[2–6] However, the practical application of PBT has some limitations, notably when using an athlete's previous maximal ability to prescribe training loads. This approach can be problematic if the athlete's 1RM changes as a consequence of training, leading to a situation where the prescribed load may not match the intended percentage of 1RM for the specific session.[7] Moreover, the precision of translation of the percentage of 1RM to different repetition maxima (e.g. 87% 1RM = 5RM) for prescription purposes may vary between individuals and exercises.[8] These issues may lead to improper loading for desired training goals, particularly if the training goal is to improve a specific segment of the individual load-velocity profile (LVP). Therefore, a methodology of training prescription more sensitive to individual true performance readiness on a given day may allow

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for more appropriate load and volume prescriptions so as to enhance specific performance.

Advances in sport technology have allowed for more objective means to monitoring an individual daily performance readiness, such as the use of linear positional transducer (e.g. GymAware, SmartCoach, T-Force, and Musclelab), laser optic technology (e.g. FLEX), or inertial measurement unit (e.g. Enode) to quantify barbell movement velocity during resistance training. Velocity-based training (VBT) is a methodology of resistance training programming that uses instantaneous velocity feedback to objectively monitor and adjust the training load and repetitions.[1] Although various methods exist within the VBT,[9] two main approaches include velocity zones and velocity loss (VL) thresholds. Velocity zones involve completing a set of repetitions at a mean concentric velocity (MCV) or mean propulsive velocity (MPV)[10] that is within a predefined threshold (e.g. 0.60–0.70 m/s). These velocity zones can be derived from an individual LVP. The LVP can be established using a progressive load–velocity test before training,[7,11] which is a crucial reference for velocity zone and load selection. In practice, a load eliciting a barbell movement within a predefined velocity zone is selected. The load may decrease per training set if movement velocity falls out of a prescribed velocity zone, allowing for a relatively more standardized training stimulus across training sets and relatively tight control over training-induced fatigue.[12]

VL thresholds involve performing repetitions within a set until the repetition velocity drops below a prescribed cutoff.[1] The mean velocity from the fastest (and typically the first) repetition of each exercise set is set as the reference,[11] and VL is calculated as [(fastest velocity - slowest velocity)/fastest velocity] × 100%.[13] After the observed VL becomes higher than the prescribed VL cutoff, the set is terminated. In each set, VL control enables a consistent control of movement velocity, power, perceived exertion rating, and blood lactate level.[12,14] Several studies have compared the training effects of low (0%–20%) and high (30%–45%) rating,[11,15–19] and their results have indicated that VL magnitude may influence subsequent training adaptations. For example, improvement in power or tasks involving high movement velocities, such as squat velocity against light load (MCV or MPV of >1 m/s), jump height, and sprint time, may be more effectively achieved using low VL training.[11,15–18] By contrast, high VL training may lead to more optimized hypertrophy and muscular endurance outcomes.[15,18] Relevant studies have been equivocal regarding a superior training effect of high and low VL for maximal strength improvement.[11,15–17]

Previous review studies have evaluated the differences in performance improvement before and after training with high or low VL,[7,20–23] as well as the dose-response relationship between the training effect and VL.[20] However, these meta-analysis studies[20–22] did not specifically compare the squat training effects between groups experiencing varying levels of VL within each included study. Therefore, this study aims to perform a meta-analysis focusing on the intergroup differences within each study included. We used meta-analysis to evaluate the differences in performance-based outcomes between resistance training with low and high VL and the potential moderating effects of different VL schemes on performance-based outcomes. Finally, this study determined the recommended VL ranges for various performance goals based on the current results. We hypothesized that compared with training with high VL, training with low VL elicits larger training effects on velocity-dependent performance outcomes and that there are optimal VL thresholds for various performance goals.

Methods

Protocol and registration

This study followed the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement for reporting systematic reviews and meta-analysis.[24] Before the start of the study, the review protocol was registered in the PROSPERO (No.: CRD42021247929).

Search strategy

We performed a comprehensive search for literature with titles and abstracts containing the relevant keywords in five electronic databases: Embase, Medline, The Cochrane Library, Scopus, and EBSCO. The search date was May 21–22, 2023. Two authors (BYC and WHC) discussed and developed a systematic search strategy (Supplementary Appendix S1) by clustering keywords according to the PICOS (patient/problem, intervention, control/comparison, outcome, and study design) strategy (Table 1). The electronic databases were mainly searched independently by the two aforementioned authors. The selected keywords included synonyms and controlled vocabulary (from MeSH and Emtree) associated with resistance training interventions and velocity variable regulation. The keywords were combined using Boolean terms (AND and OR) and the randomized controlled trial filter[25] was used to select randomized controlled trials alone. A detailed list of the exact terms and search strategies used is presented in Supplementary Appendix S1. The reference lists in selected studies were manually searched for additional eligible studies.

Inclusion criteria

The PICOS method[25] was used to set criteria for study inclusion (Table 1).

Training protocols

We divided all training protocols in the included literature into two training subgroups—VL0–20 (VL threshold = 0%–20%) and VL30–45 (VL threshold = 30%–45%).

Table 1. PICOS.

Items	Definition
Problems	The chronic effect of squat training with different VL thresholds on strength, power, and speed performance.
Participants	1. Healthy people. 2. Male or female. 3. Non-pathological. 4. Resistance training experience.
Intervention	1. Resistance training programs included back-squat exercise. 2. Using technology instrument (e.g. IMUs, LPTs, etc.) to monitor movement velocity over all training. 3. The set was terminated at prescribed percentage VL thresholds. 4. Velocity variables presented in studies: mean concentric velocity, mean propulsive velocity, and percentage velocity loss. 5. Training period at least 4 weeks. 6. Excluding in-season training. 7. Excluding concurrent training.
Control/ compare	Training with VL threshold
Outcomes	1. Performance outcomes (at least one outcome): (a) One-repetition maximum squat strength. (b) Heavy-load squat velocity: MCV or MPV attained against an absolute load common to pre- and post-test that was moved slower than 1 m/s. (c) Light-load squat velocity: MCV or MPV attained against an absolute load common to pre- and post-test that was moved faster than 1 m/s. (d) Countermovement jump height. (e) 20-meter sprint time. 2. Statistical outcomes (at least one outcome): (a) Means and standard deviations (standard errors) data. (b) Effect size (standard mean difference) and confidence interval data. (c) *p*-Value data were provided in included studies outcome measures.
Study design	Peer-reviewed article and randomized control trial.

VL: velocity loss; IMU: inertial measurement unit; LPT: linear position transducer; MCV: mean concentric velocity; MPV: mean propulsive velocity.

Next, we compared the pretraining and posttraining differences within each subgroup as well as between VL0–10 and VL20 and between VL0–20 and VL30–40.

Outcome measures

Five outcome measures were included in this study: (a) 1RM squat strength, (b) heavy-load squat velocity (HLSV; MCV or MPV attained against an absolute load, common in pretraining and posttraining, that was moved slower than 1 m/s),[11] (c) light-load squat velocity (LLSV; MCV or MPV attained against an absolute load, common in pretraining and posttraining, that was moved faster than 1 m/s),[11] (d) countermovement jump (CMJ) height, and (e) 20-m sprint time. If some studies contributed >1 effect size (ES) for each outcome (e.g. HLSV at different loads and sprint time at different distances), we combined the results to avoid double-counting and correlated ESs.[25]

Data extraction

Two authors (BYC and WHC) independently extracted data from the included studies. Disagreements concerning the selection criteria were discussed and resolved through the consensus of all authors (BYC, WHC, NJF, CFH, PF, and TYS). In cases where studies were acceptable based on the inclusion and exclusion criteria, but they did not report preintervention and postintervention data or intervention ES, the corresponding authors or co-authors of the studies were contacted by email (BYC).

Methodological quality

Two authors (BYC and WHC) independently used the Cochrane risk of bias tool[26] to analyze the quality of research methods. This tool evaluates the (a) randomness of the allocation sequence (selection bias), (b) concealment of the allocation sequence (selection bias), (c) blinding of participants and personnel (performance bias), (d) blinding of outcome assessment (detection bias), (e) incomplete outcome data (attrition bias), (f) selective outcome reporting (reporting bias), and (g) any other biases. Each study was labeled as low, high, or unclear risk of bias.

Statistical analyses

The meta-analysis and statistical analysis were performed using Comprehensive Meta-Analysis (version 3; Biostat, Englewood, NJ, USA) to calculate continuous data as Hedges' g ES and its 95% confidence intervals[27] and integrate the results and create forest plots. The combined ES pretraining and posttraining was calculated to investigate the training effect for each training subgroup. The combined ES was calculated for the intervention and control groups in each study to investigate the differences between different trainings. Statistical significance was set at $p<0.05$. According to Cohen's prescription, trivial, small, moderate, and large ESs are defined as $\leq$0.2, $\leq$0.5, $\leq$0.8, and $\geq$0.8, respectively.[28] The random-effects model was used to combine ES when heterogeneity existed between published studies.[29] The weight of each study was calculated using the reciprocal of within-study variance and between-study variance tau-square (τ^2).[30] Low, medium, and high between-study heterogeneity was defined

by an I^2 of 25%, 50%, and 75%, respectively.[31] The trim and fill method was used to examine the asymmetry of the funnel plot.[32,33] We performed a sensitivity analysis by using the asymmetry results to determine the stability of the combined results. We also performed meta-regression to analyze the interaction between training effects and VL to predict training effects based on a single independent variable (i.e. VL). Considering the potential heterogeneity between different periodization trainings, only linear programming (linear progressive training) studies were analyzed in this meta-regression because studies on other training periodizations ($k<5$) are scant.

Results

Study selection

Our electronic database search yielded 1017 articles ($n=$ 107, 330, 122, 239, and 219 from Medline, Embase, The Cochrane Library, Scopus, and EBSCO, respectively). After duplicates were removed, the titles and abstracts were assessed, and full-texts were screened, only 13 studies,[11,15–17,19,34–41] which met the inclusion criteria, were included in the meta-analysis (Figure 1, Supplementary Appendix S2).

Study characteristics

The pooled number of participants across all included studies was 308. Sample sizes in individual experimental groups ranged from 8 to 16. All studies included participants with previous resistance training experience (>2 years in four studies, >1.5 years in four studies, >1 year in four, and no description in a study). The participants' average age ranged from 14 to 24 years and squat 1RM ranged from 1.25 to 1.8 body weight across all studies. The training program durations ranged from 6 to 8 weeks, with the external load lifted corresponding to an intensity of 50% to 95% 1RM. The VL threshold ranged from 0% to 45%. All the 13 included studies assessed the maximum strength using 1RM tests, whereas several studies assessed it using the HLSV ($n=9$) and LLSV ($n=8$), CMJ height ($n=11$), and sprint time ($n=6$). Table 2 and Supplementary Appendix S3 provide the characteristics and outcome data of the included studies, respectively.

Bias assessment

In the bias assessment, 15.4% of the studies demonstrated high risks of incomplete outcome data and other bias. All the included studies (100%) demonstrated an unclear risk of allocation concealment selection bias item. The risk of bias assessment summary and the percentage of each risk of bias item across all the included studies are presented in Figure 2. The detailed descriptions and evaluation evidence for each study are provided in Supplementary Appendix S4. The trim and fill results were asymmetric in VL0–20 (HLSV) and VL30–40 (1RM); the relevant

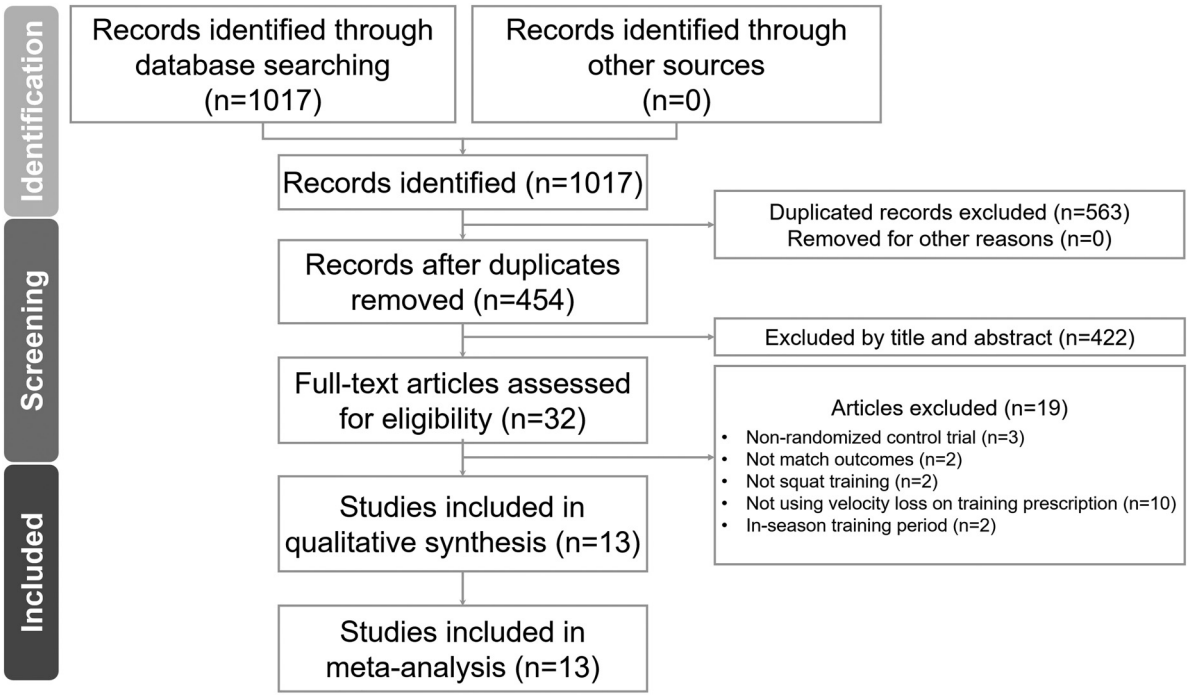

Figure 1. PRISMA flow diagram.

Table 2. Study characteristics for groups, participants, training program, and outcomes.

Study	Group	Participant	Training program	Measurement tool	Outcome
Costilla et al. (2023)	VL20	8 M (age: 20.4 ± 2.2) TE: > 2 years; RS: 1.46 BW	6 weeks LP SQT Target velocity refer to 70–80% 1RM—5 sets*VL20%	Smart Coach (linear positional transducer)	1RM HLSV
	VL40	7 M (age: 22.3 ± 2.6) TE: > 2 years; RS: 1.34 BW	6 weeks LP SQT Target velocity refer to 70–80% 1RM—5 sets*VL40%		
Dorrell et al. (2020)	VL20	8 M (age: 22.8 ± 4.5) TE: > 2 years; RS: 1.54 BW	6 weeks LP SQT and ADT TV refer to 70–95% 1RM—3 sets*VL20%	GymAware (linear positional transducer)	1RM CMJ height
Galiano et al. (2022)	VL5	15 M (age: 22.1 ± 2.9) TE: > 1.5 years; RS: 1.34 BW	7 weeks CP SQT TV 1.14 m/s (~50% 1RM)—3 sets*VL5%	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height 20 m ST
	VL20	13 M (age: 23.9 ± 3.0) TE: > 1.5 years; RS: 1.29 BW	7 weeks CP SQT TV 1.14 m/s (~50% 1RM)—3 sets*VL20%		
García-Orea et al. (2023)	VL15	9 M (age: 23.9 ± 5.3) TE: no description; RS: 1.28 BW	6 weeks LP SQT Target velocity 1.07–0.84 m/s (55–70% 1RM)—3 sets*VL15% (17.3±1.9)	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height
Held et al. (2021)	VL10	11 B (age: 19.8 ± 2.3) TE: > 2 years; RS: 1.63 BW	8 weeks CP SQT and ADT TV 0.67 m/s (80% 1RM)—4 sets* VL10%	Speed4lifts encoder (linear positional transducer)	1RM
Myrholt et al. (2023)	VL20	5M3F (age: 26.6 ± 3.5) TE: > 1 years; RS: 1.3–1.8 BW	6 weeks UP SQT (equalize the training volume between VL20 and VL40) Target velocity 0.70–0.49 m/s (79–89% 1RM)—mutiple sets*VL20% (21.0±1.0)	Musclelab (linear positional transducer)	1RM CMJ height
	VL40	4M4F (age: 27.8 ± 3.6) TE: > 1 years; RS: 1.3–1.8 BW	6 weeks UP SQT (equalize the training volume between VL20 and VL40) Target velocity 0.70–0.49 m/s (79–89% 1RM)—mutiple sets*VL40% (40.0±2.0)		
Pareja-Blanco et al. (2017)	VL20	12 M (age: 22.7 ± 1.9) TE: 1.5–4 years; RS: 1.41 BW	8 weeks LP SQT TV refer to 70–85% 1RM—3 sets*VL20%	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height 20 m ST
	VL40	12 M (age: 22.7 ± 1.9) TE: 1.5–4 years; RS: 1.41 BW	8 weeks LP SQT TV refer to 70–85% 1RM—3 sets*VL40%		
Pareja-Blanco et al. (2020a)	VL0	14 M (age: 24.1 ± 4.3) TE: 1.5–4 years; RS: 1.32 BW	8 weeks LP SQT TV refer to 70%–85% 1RM—3 sets*VL0% (1 rep)	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height 20 m ST
	VL10	14 M (age: 24.1 ± 4.3)			

(continued)

Table 2. (continued)

Study	Group	Participant	Training program	Measurement tool	Outcome
		TE: 1.5–4 years; RS: 1.28 BW	8 weeks LP SQT TV refer to 70%–85% 1RM—3 sets*VL10%		
	VL20	13 M (age: 24.1 ± 4.3) TE: 1.5–4 years; RS: 1.30 BW	8 weeks LP SQT TV refer to 70%–85% 1RM—3 sets*VL20%		
	VL40	14 M (age: 24.1 ± 4.3) TE: 1.5–4 years; RS: 1.25 BW	8 weeks LP SQT TV refer to 70%–85% 1RM—3 sets*VL40%		
Rebelo et al. (2023)	VL20	9F (age, 14.0 ± 1.1) TE: > 2 years; RS: 1.40 BW	6 weeks UP SQT and ADT Target velocity refer to 85–90% 1RM—3 sets*VL20%	FLEX (laser optic technology)	1RM CMJ height
Riscart-Lopez et al. (2021)	VL20	11 M (age: 23.5 ± 6.3) TE: 1.5–4 years; RS: 1.3 BW	8 weeks LP SQT TV refer to 50–85% 1RM—3 sets*VL20%	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height 20 m ST
	VL20	10 M (age: 23.5 ± 5.2) TE: 1.5–4 years; RS: 1.31 BW	8 weeks UP SQT TV refer to 50–85% 1RM—3 sets*VL20%		
	VL20	11 M (age: 21.9 ± 2.9) TE: 1.5–4 years; RS: 1.41 BW	8 weeks RP SQT TV refer to 50–85% 1RM—3 sets*VL20%		
	VL20	11 M (age: 22.5 ± 4.7) TE: 1.5–4 years; RS: 1.27 BW	8 weeks CP SQT TV refer to 67.5% 1RM—3 sets*VL20%		
Rodriguez-Rosell et al. (2020)	VL10	12 M (age: 22.8 ± 3.1) TE: 1–3 years; RS: 1.34 BW	8 weeks LP SQT TV 0.84–0.60 m/s—3 sets*VL10%	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height 10 m, 20 m ST
	VL30	13 M (age: 22.2 ± 2.7) TE: 1–3 years; RS: 1.31 BW	8 weeks LP SQT TV 0.84–0.60 m/s—3 sets*VL30%		
Rodriguez-Rosell et al. (2021a)	VL15	16 M (age: 24.3 ± 4.2) TE: 1–3 years; RS: 1.33 BW	8 weeks LP SQT TV 1.16–0.68 m/s (50–80% 1RM)—3 sets*VL15%	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height
	VL15	16 M (age: 22.4 ± 2.7) TE: 1–3 years; RS: 1.33 BW	8 weeks UP SQT TV 1.16–0.68 m/s (50–80% 1RM)—3 sets*VL15%		
Rodriguez-Rosell et al. (2021b)	VL10	11 M (age: 22.8 ± 3.9) TE: 1–3 years; RS: 1.36 BW	8 weeks LP SQT Target velocity 1.08–0.84 m/s (55–70% 1RM)—3 sets*VL10% (10.9±1.8)	T-Force (linear positional transducer)	1RM HLSV, LLSV CMJ height
	VL30	11 M (age: 21.9 ± 2.3)	8 weeks LP SQT		

(continued)

Table 2. (continued)

Study	Group	Participant	Training program	Measurement tool	Outcome
		TE: 1–3 years; RS: 1.32 BW	Target velocity 1.08–0.84 m/s (55–70% 1RM)—3 sets*VL30% (29.8±3.6)		10 m, 20 m ST
	VL45	11 M (age: 21.6 ± 2.8) TE: 1–3 years; RS: 1.34 BW	8 weeks LP SQT Target velocity 1.08–0.84 m/s (55–70% 1RM)—3 sets*VL45% (42.1±7.0)		

VL: velocity loss; M: male; F: female; B: both male and female; TE: training experience; RS: relative strength of 1RM; BW: body weight; SQT: squat training; ADT: additional training; LP: linear programming; UP: undulating programming; RP: reverse programming; CP: constant programming; TV: target velocity; 1RM: one-repetition maximum squat strength; HLSV: heavy-load squat velocity (mean propulsive velocity attained against an absolute load common to pre- and post-test that was moved slower than 1 m/s); LLSV: light-load squat velocity (mean propulsive velocity attained against an absolute load common to pre- and post-test that was moved faster than 1 m/s); CMJ: countermovement jump; ST: sprint time.

funnel plots are presented in Supplementary Appendix S5. After sensitivity analysis, all filled results showed stable combined results (Supplementary Appendix S6).

Meta-analysis

Pretraining and posttraining effects. Two training protocols (VL0–20 and VL30–45) were assessed for their pretraining and posttraining effects on 1RM, HLSV, LLSV, CMJ height, and sprint time (Figures 3–7). All outcomes demonstrated no heterogeneity ($I^2 = 0\%$), except to HLSV (VL0–20: $I^2 = 62.952\%$, $p = 0.000$; VL30–45: $I^2 = 49.439\%$, $p = 0.078$). Both the training protocols significantly improved 1RM (ES = 0.69, $p < 0.001$; Figure 3), HLSV (ES = 1.59–2.09, $p < 0.001$; Figure 4), LLSV (VL0–20:ES = 0.98, $p < 0.001$; VL30–40: ES = 0.59, $p = 0.001$; Figure 5), and CMJ height (VL0–20: ES = 0.65, $p < 0.001$; VL30–40: ES = 0.438, $p = 0.009$; Figure 6). By contrast, as shown in Figure 7, VL0–20 protocols significantly improved sprint time after training (ES = 0.59, $p < 0.001$) but VL30–45 protocol led to no significant pretraining–posttraining differences (ES = −0.013, $p = 0.949$).

Low VL vs. high VL. No heterogeneity was observed among the included studies ($I^2 = 0\%$) for all low and high VL outcomes (Figures 8 and 9), except to HLSV ($I^2 = 1.152\%$, $p = 0.420$) and LLSV ($I^2 = 26.396\%$, $p = 0.227$) between VL0–20 and VL30–45. No significant differences were noted between VL0–10 and VL20 in terms of 1RM (ES = −0.02, $p = 0.940$), HLSV (ES = 0.23, $p = 0.278$), LLSV (ES = 0.15, $p = 0.493$), CMJ height (ES = 0.16, $p = 0.454$), and sprint time (ES = −0.31, $p = 0.403$; Figure 8). Similarly, no significant differences were noted between VL0–20 and VL30–45 in terms of 1RM (ES = −0.65, $p = 0.630$) and CMJ (ES = −0.24, $p = 0.088$) between VL0–20 and VL30–45; by contrast, HLSV (ES = −0.40, $p = 0.005$), LLSV (ES = −0.39, $p = 0.025$), and sprint time (ES = −0.47, $p = 0.022$) significantly differed between VL0–20 and VL30–45 (Figure 9).

Meta-regression

Regarding the pretraining and posttraining effects of LLSV (coefficient = -0.0142, 95%CI [-0.03, 0.00], $p = 0.1039$) and CMJ (coefficient = -0.0151, 95%CI [-0.03, 0.00], $p = 0.0757$) performance (Figure 10(a)), the interaction was nearly significant and negative in the range of 0% to 45% VL. Regarding the pretraining and posttraining effect of sprint time performance (Figure 10(b)), the interaction was significant and negative (coefficient = -0.0317, 95% CI [-0.05, -0.01], $p = 0.002$) in the range of 10% to 40% VL. No interactions were observed for 1RM ($p = 0.550$) and HLSV ($p = 0.259$).

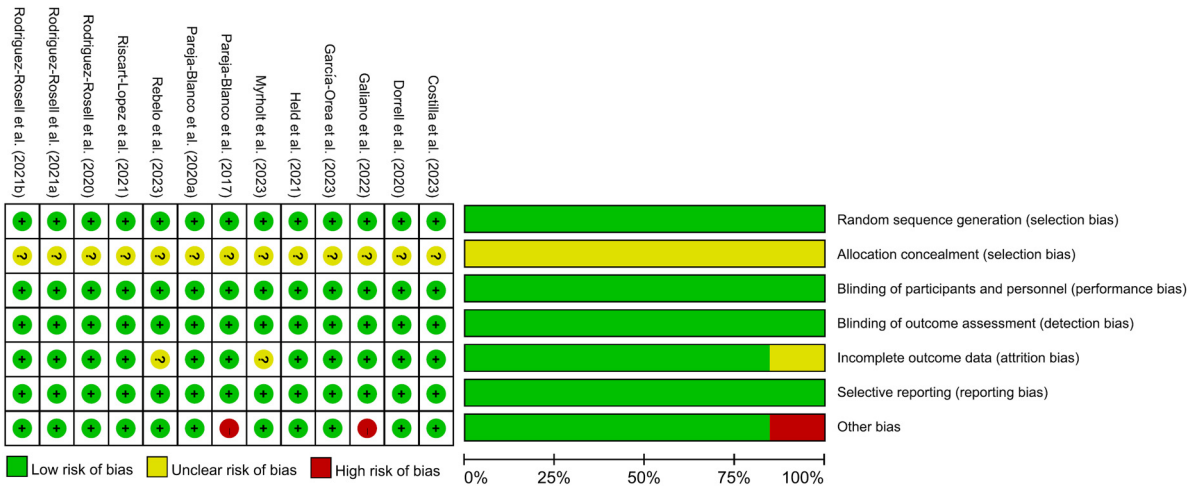

Figure 2. Risk of bias assessment summary.

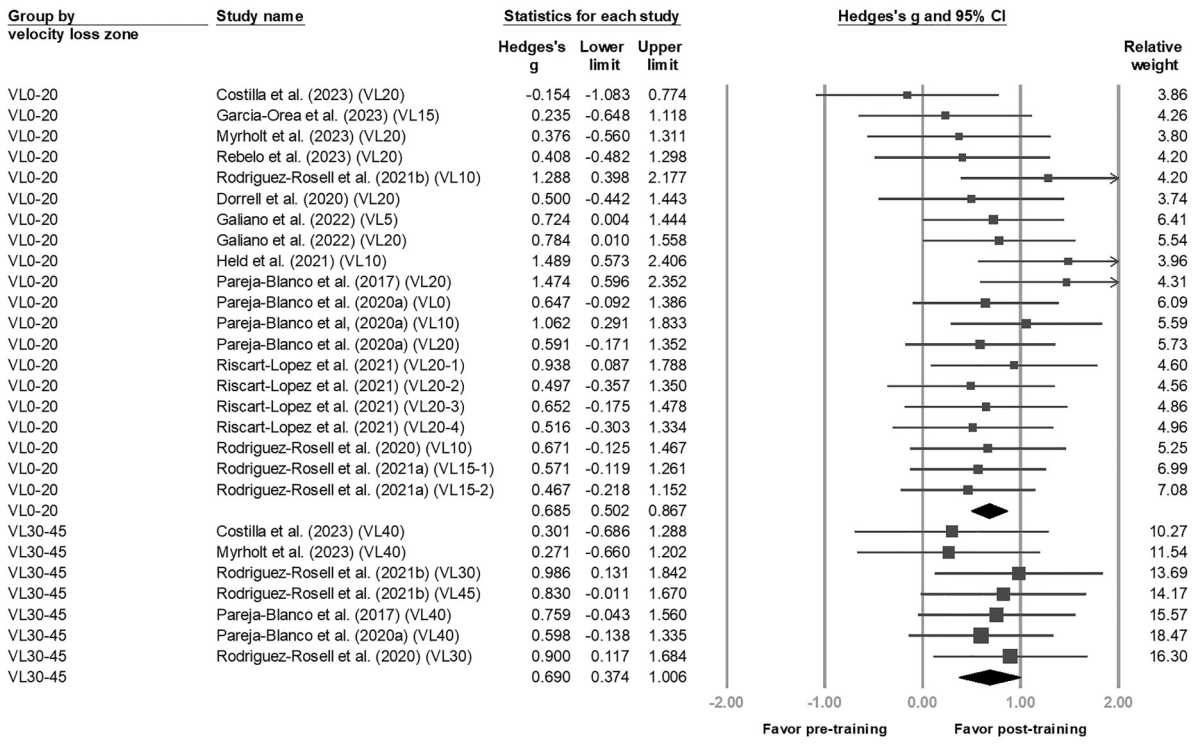

Figure 3. Effect of squat resistance training with velocity loss threshold on one-repetition maximum squat strength. VL0–20 and VL30–45 = velocity loss thresholds in the range of 0%–20% and 30%–40%, respectively. The sizes of the squares denote the study weights.

Discussion

This study used a meta-analysis and meta-regression to systematically evaluate the short-term effects of squat resistance training with different VL thresholds on squat strength, HLSV, LLSV, CMJ height, and sprint time. Results revealed that squat resistance training with a low (0%–20%) or high (30%–45%) VL scheme improved the squat strength, HLSV, LLSV, and CMJ height but that only squat resistance training with a low VL scheme improved sprint time after short-term training. The differences between the outcomes of low (0%–20%) and high (30%–45%) VL training was significant in terms of HLSV, LLSV, CMJ height ($p = 0.088$) and sprint time. Moreover, the interaction effects were significant between VL threshold and CMJ height ($p = 0.076$) and

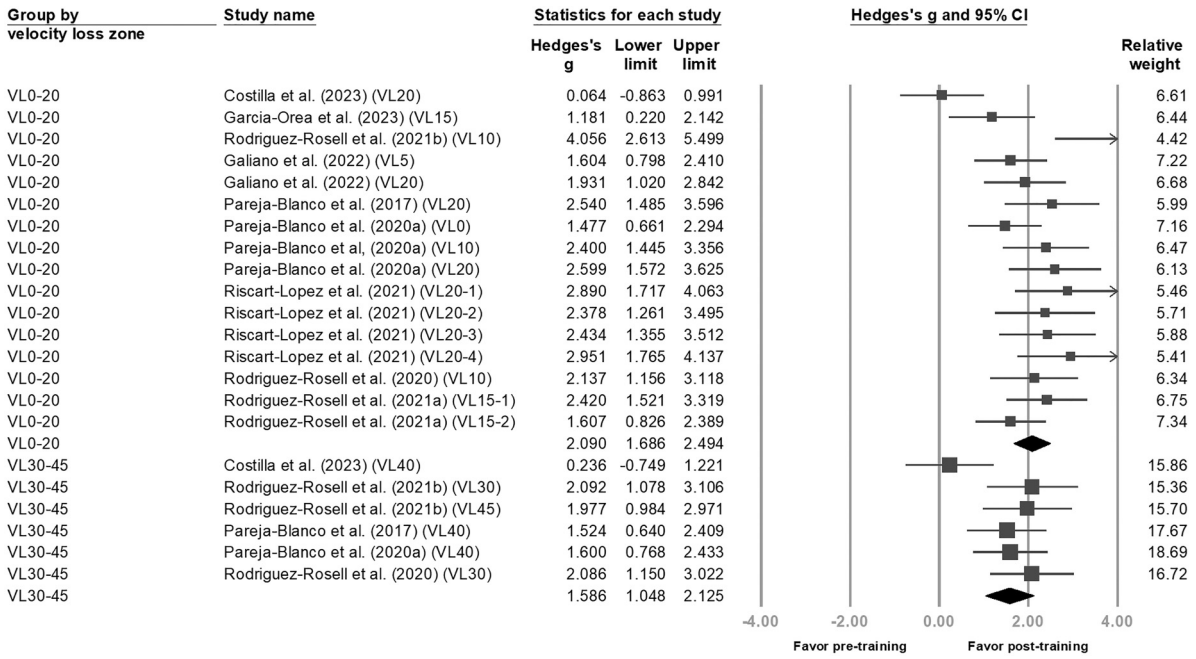

Figure 4. Effect of squat resistance training with velocity loss threshold on heavy-load squat velocity. VL0–20 and VL30–45 = velocity loss thresholds in the range of 0%–20% and 30%–40%, respectively. The sizes of the squares denote the study weights.

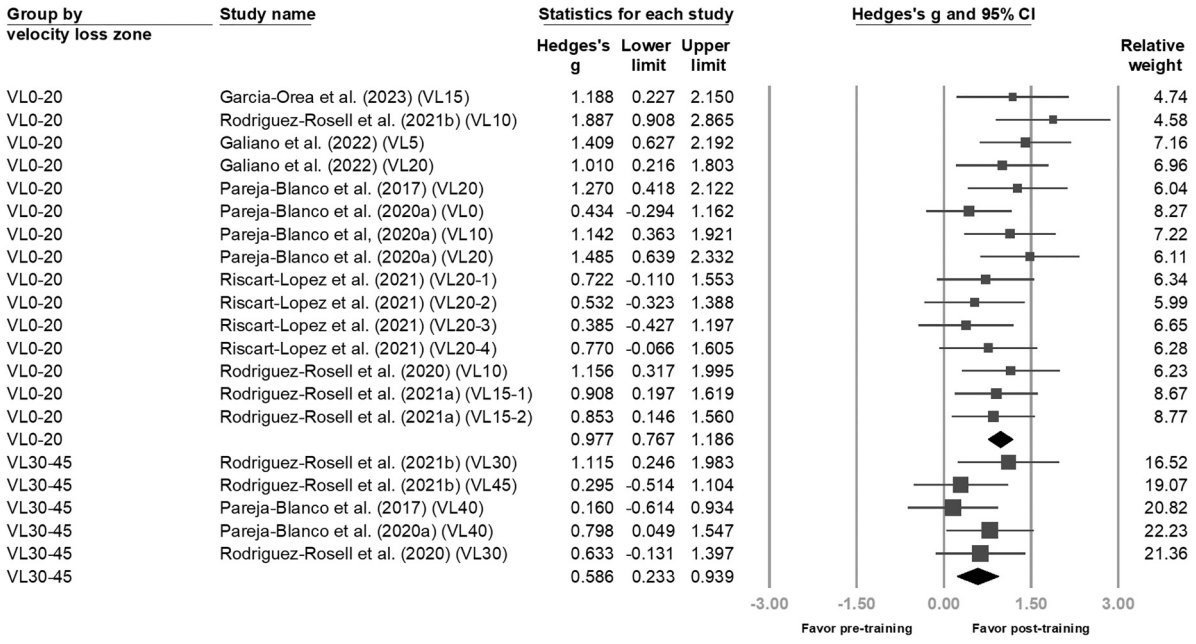

Figure 5. Effect of squat resistance training with velocity loss threshold on light-load squat velocity. VL0–20 and VL30–45 = velocity loss thresholds in the range of 0%–20% and 30%–40%, respectively. The sizes of the squares denote the study weights.

sprint time ($p = 0.010$)—indicating that performance enhancement was possibly dependent on VL magnitude. Taken together, these results revealed that squat resistance training with low (0%–20%) VL led to more improvement in power and speed tasks than did that with high (30%–45%) VL.

Our meta-analysis showed that different VL schemes have no significant influence on squat strength changes (Figures 8 and 9)—indicating that high VL is not a necessary condition for improving maximal strength. Such result also demonstrated in previous study.[11,15–17] Training to a high performance fatigability, such as training

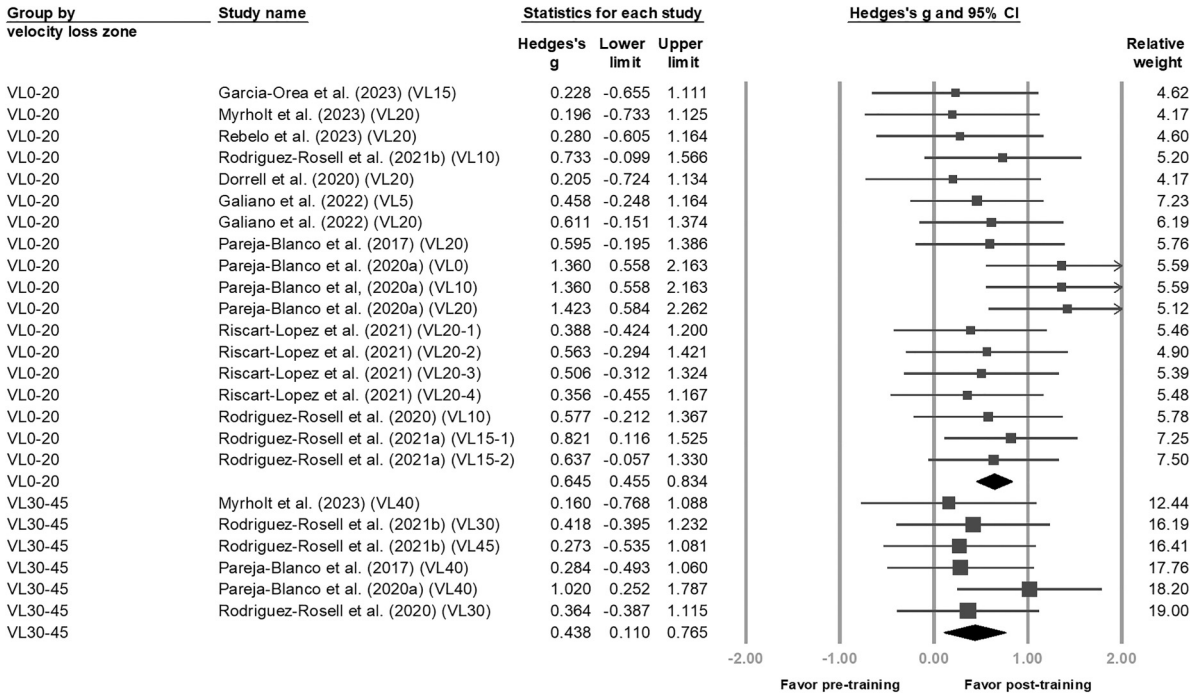

Figure 6. Effect of squat resistance training with velocity loss threshold on countermovement jump height. VL0–20 and VL30–45 = velocity loss thresholds in the range of 0%–20% and 30%–45%, respectively. The sizes of the squares denote the study weights.

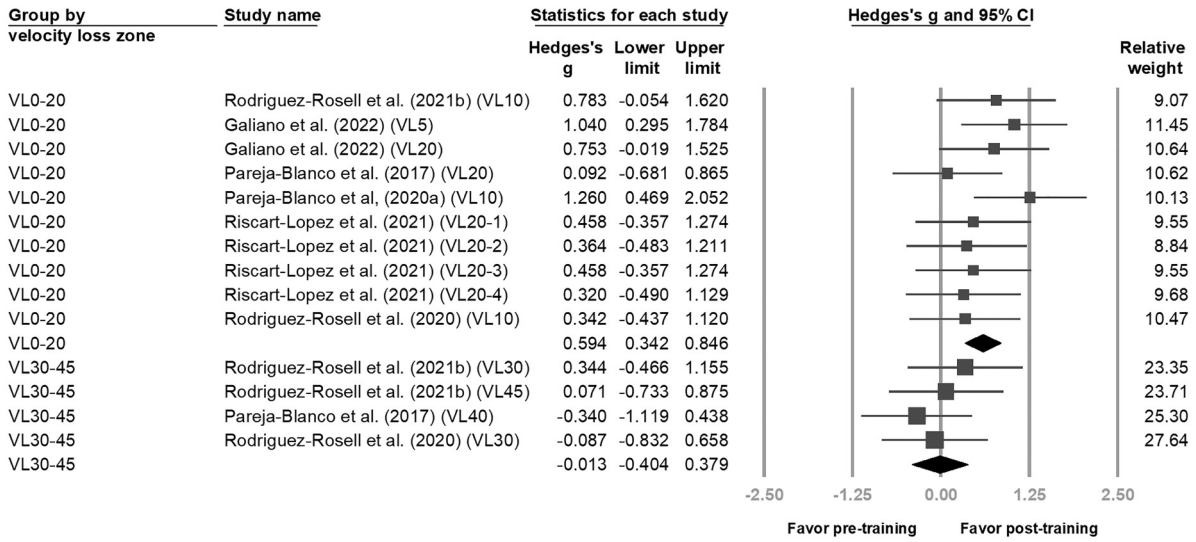

Figure 7. Effect of squat resistance training with velocity loss threshold on sprint time. VL0–20 and VL30–45 = velocity loss thresholds in the range of 0%–20% and 30%–45%, respectively. The sizes of the squares denote the study weights.

to failure, has been reported that not provide training effects superior to those of nonfailure training.[42] Thus, maximum strength can be increased if the load is appropriate to induce an overload stimulus and the training volume (sets × repetitions) reaches a certain level.[43]

Our results indicated that training with a low VL (0%–20%) had more favorable effects on HLSV (ES = -0.40, $p = 0.005$) and LLSV (ES = -0.39, $p = 0.025$) compared to high VL (30%–45%) (Figure 9). Furthermore, the benefit on HLSV and LLSV of using a VL20 showed a non-significant trend over the range of VL0–10 (HLSV: ES = 0.234, $p = 0.278$; LLSV: ES = 0.15, $p = 0.493$, Figure 8). In our meta-regression analysis, we did not observe any significant interaction effect for training effects on HLSV and

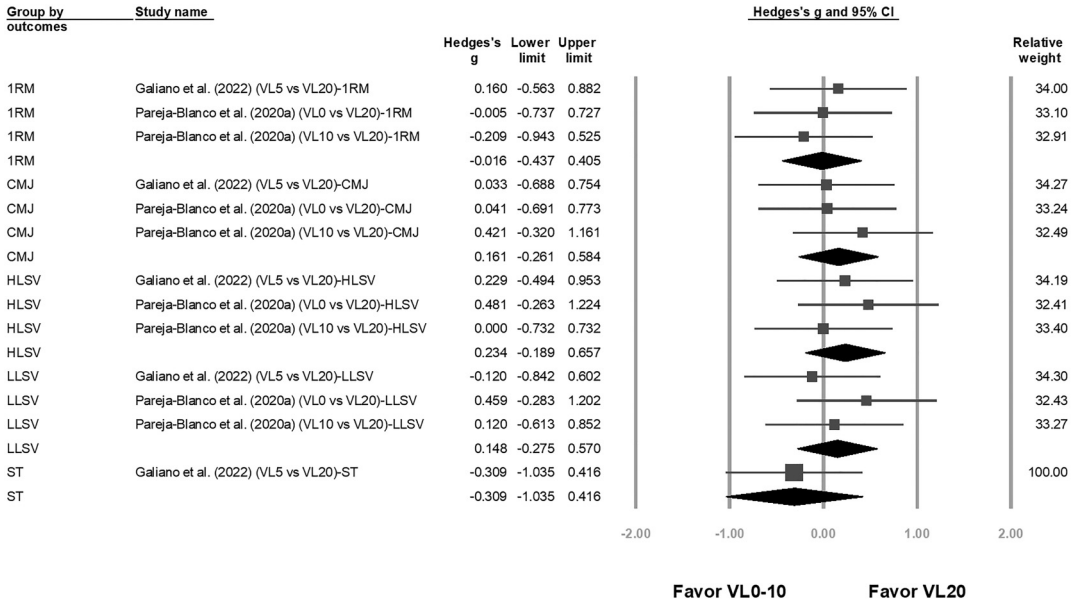

Figure 8. Comparison of squat resistance training with VL0–10 and VL20 on strength, power, and speed performance. 1RM = one-repetition maximum squat strength; CMJ = countermovement jump; HLSV = heavy-load squat velocity; LLSV: light-load squat velocity; VL0–10 and VL20 = velocity loss thresholds in the range of 0%–10% and at 20%, respectively. The sizes of the squares denote the study weights.

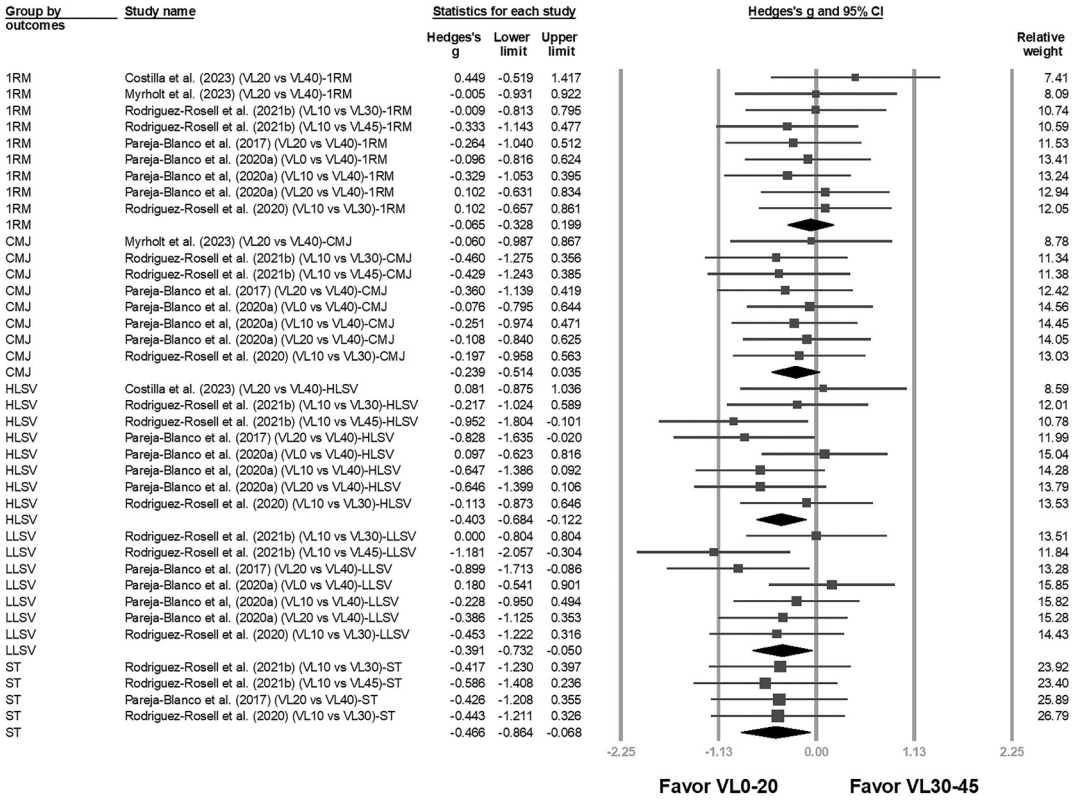

Figure 9. Comparison of squat resistance training with VL0–20 and VL30–45 on strength, power, and speed performance. 1RM = one-repetition maximum squat strength; CMJ = countermovement jump; HLSV = heavy-load squat velocity; LLSV = light-load squat velocity; VL0–20 and VL30–45 = velocity loss thresholds in the range of 0%–20% and 30%–45%, respectively. The sizes of the squares denote the study weights.

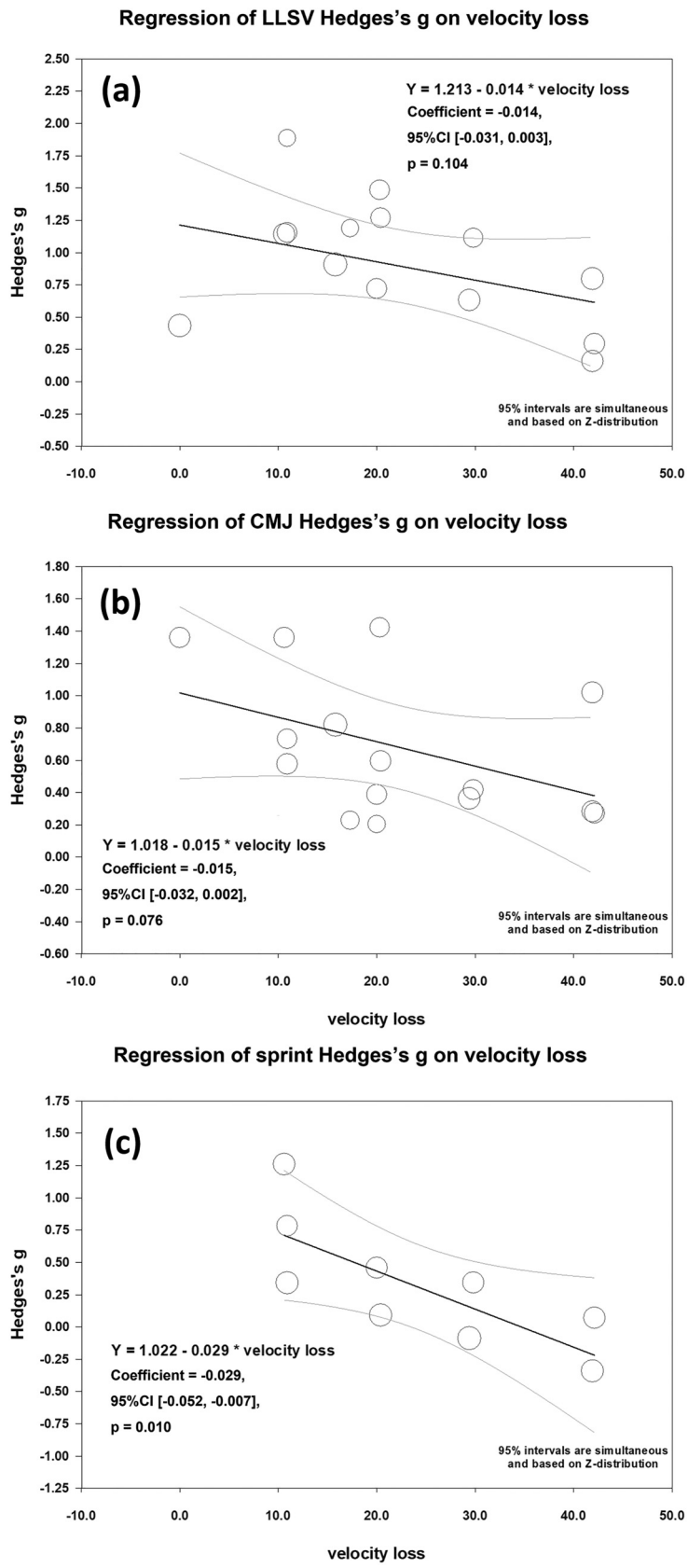

Figure 10. Meta-regression of velocity loss and training effects on (A) HLSV, (B) LLSV, and (C) sprint time. HLSV = heavy-load squat velocity; LLSV = light-load squat velocity. The sizes of hollow circles denote the study weights.

LLSV within the range of VL 0–45 (Figure 10(a)). However, we did notice a linear decrease in training effects on LLSV performance with increasing VL beyond 20%, although this decrease did not reach statistical significance. As these results, it appears that resistance training with a VL in the range of 10% to 20% may be more optimal for improving HLSV and LLSV.

For the measurement of CMJ height, the meta-analysis results did not reveal a statistically significant difference between low VL (0%–20%) and high VL (30%–45%) (ES = -0.24, $p = 0.088$, as shown in Figure 9). However, it is worth noting that low VL appeared to have more favorable training effects (Figures 9 and 10(b)). Even though resistance training with both low (0%–20%) and high VL (30%–45%) resulted in similar improvements in CMJ, the total training volume differed significantly.[11,15] The lower VL thresholds is more efficient, and they do not cause high levels of discomfort and fatigue typically experienced in training routines characterized by high VL thresholds.[44] Additionally, maintaining a higher movement velocity and avoiding excessive performance fatigability during training may be beneficial for improving CMJ height.[11] Therefore, in practical applications, we would recommend utilizing a VL of 20% or lower to maintain a higher movement velocity and ensure time efficiency.

Our meta-analysis result demonstrated that sprint time significantly improved after training with VL0–20 ($p < 0.05$, ES = 0.59) but not VL30–45 ($p > 0.05$, ES = −0.01, Figure 7). Moreover, VL0–20 was confirmed to be superior to VL30–45 ($p < 0.05$, ES = −0.47, Figure 9). The meta-regression analysis also indicated a significant negative interaction effect ($p = 0.010$, Figure 10(c)) between training effect and VL. These results indicated that sprint performance improvement may be optimized using high velocity with respect to the lifted load during training[45] and with tight control over VL such that each repetition velocity is sufficient.[15] Greater neuromuscular fatigue associated with high VL[13] may negatively affect sprinting performance[46] and explain the observed maladaptation. Moreover, sprint performance is influenced by type IIX muscle fiber composition.[47,48] Pareja-Blanco and Rodríguez-Rosell[11] found that the type IIX muscle fiber composition reduced after 8 weeks of training with high (40%) VL but not with low (20%) VL. Therefore, both fatigue and muscle fiber shifts may be the mechanisms underlying the high VL maladaptation. In summary, if the purpose of training is to improve sprint performance, maintaining a high movement velocity with maximal intent with respect to the lifted load and minimizing VL magnitude may produce optimized training effects.

For practical recommendations, if the training target is to improve strength, power, and speed performance, squat resistance training with VL ≤ 20% is recommended. Moreover, if the purpose of training is to enhance maximum squat strength alone, tight VL control may not be necessary; a VL range of 0% to 40% has similar improvements. However, if the purpose is to improve HLSV or LLSV, a VL of approximately 10% to 20% is recommended. A VL of <20% and ≤10% may be preferable if the immediate goal is to optimize CMJ height and sprint performance, respectively. The VL ranges in squat training recommended for different performance goals are presented in Table 3.

Table 3. Recommended VL ranges for squat training by performance goal.

Performance goal	Squat VL range
Maximum squat strength	0–40%
Heavy-load squat velocity	10–20%
Light-load squat velocity	10–20%
CMJ height	≤20%
Sprint time	≤10%

VL: velocity loss. Heavy-load squat velocity: concentric/propulsive velocity against a load that was moved slower than 1 m/s. Light-load squat velocity: concentric/propulsive velocity against a load that was moved faster than 1 m/s. CMJ: countermovement jump.

The conclusions of the current study are limited with respect to the methodologies employed in the evaluated studies. The experimental nature of the studies may affect the ecological considerations, particularly in terms of long-term training for performance optimization. First, the included studies were only 6 to 8 weeks long and comprised limited exercise selection (i.e. squat), limited programming (i.e. set-rep schemes, loading ranges, tapering, etc.), and limited periodization (i.e. distinct and systematic fitness phases). In practice, an athlete's training cycle may span many months and include long-term variation in the emphasized fitness characteristics and subsequent programming and exercise selections. Moreover, the analyzed studies did not include training of two of the criterion tasks: CMJ and sprinting. In practice, athletes are unlikely to perform resistance training completely independent of sport-specific practice. The integration of additional training may affect the magnitude of change or necessitate modifications to resistance training (e.g. athlete fatigue management).

Conclusion

The VL threshold during squat resistance training is a critical determinant for power-based performance optimization. Generally, squat resistance training with low VL threshold (≤20%) provides additional effective and efficient training stimuli for improving performance in power and speed tasks. However, optimized VL ranges differed between specific performance goals. The potentially optimal VL ranges are 10% to 20%, ≤20%, and ≤10% for HLSV and LLSV, CMJ height, and sprint time, respectively.

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Supplemental material

Supplemental material for this article is available online.

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